TAO JIANG

810 Dexter Ave N, Seattle, WA 98109
taojiang2293@gmail.com

RESEARCH INTERESTS

- **Efficiency by optimizer design:** developing optimization algorithms and theories to improve efficiency in both training and inference for large language model pretraining.
- **AI for science and science for AI:** exploring similarities between AI models and human brains and using these similarities to improve deep-learning model efficiency and performance.

EMPLOYMENT

Postdoctoral Researcher *July 2025 - Present*
FAIR, Meta Superintelligence Labs, Seattle, WA
- Develop optimization theory and algorithms to advance training and inference efficiency.
- Explore how brain data inform model pruning.

EDUCATION

Cornell University, Ithaca, NY *September 2020 - May 2025*
Ph.D. in Operations Research, School of Operations Research and Information Engineering
CGPA: 4.096/4.0
- Thesis advisor: Damek Davis
- Committee: James Renegar, Adrian Lewis, Anil Damle, Katya Scheinberg
- Thesis: Combatting Ill-Conditioning and Heavy-Tailed Noise in Non-Convex Optimization

University of Waterloo (UW), Waterloo, ON *September 2018 - August 2020*
Master of Mathematics (Thesis), Department of Combinatorics and Optimization
CGPA: 95/100
- Thesis advisor: Stephen Vavasis
- Committee: Levent Tunçel, Henry Wolkowicz
- Thesis: Sum-of-norms clustering: theoretical guarantee and algorithms

Singapore University of Technology and Design (SUTD), Singapore *May 2015 - August 2018*
Bachelor of Engineering, Summa Cum Laude, Engineering Systems and Design
CGPA: 4.79/5.00

RESEARCH INTERNSHIPS

Research Scientist Intern, FAIR Labs, Meta AI, United States *May - August 2024*
- Mentor: Sho Yaida
- Project: batch size selection and scaling for LLM pretraining.

Research Scientist Intern, FAIR Labs, Meta AI, United States *June - December 2023*
- Mentor: Lin Xiao
- Project: statistically efficient optimizers combatting heavy-tailed noise.

PUBLICATIONS AND WORKING PAPERS

I. Optimizers for deep learning

Jiang, T., Yang, T., Bu, Z., Chi, Y., Cutkosky, A., Defazio, A., Kallusky, D., & Xiao, L. (2026). Optimizer Design via Fixed-Point Iterations. (Working paper)
We propose a unified framework for momentum using fixed-point iterations. It casts LaProp, DiLoCo,

GPA, and SNOO as special cases. Based on this, we develop M2, a new optimizer that outperforms existing methods in LLM pretraining.

Golobokov, K., Jiang, T., & Xiao, L. (2026). Aggregate Proximal Method for Structured Stochastic Optimization: Convergence and Manifold Identification. (Working paper)
We prove theoretically that the AProx algorithm identifies the correct model parameters to prune and provide a convergence analysis.

Jiang, T., & Xiao, L. (2026). Second-Moment Stochastic Approximation Methods. arXiv preprint arXiv:2609.36600.
We introduce a second-moment-based paradigm for stochastic approximation methods, interpreting Adam and Muon through the same statistical lens. We develop a universal convergence theory for all algorithms in this framework and provide practical design guidelines.

Yang, T., Jiang, T., Chi, Y., Cutkosky, A., & Xiao, L. (2026). Stochastic Inertial Krasnosel'skiĭ–Mann Iteration Achieves Near-Optimal Sample Complexity. arXiv preprint arXiv:2609.28543.
We propose a hyperparameter schedule for a single-loop stochastic inertial Krasnosel'skiĭ–Mann method and prove that it achieves near-optimal $\widetilde{O}(\epsilon^{-2})$ sample complexity without batching or variance reduction.

Jiang, T., & Xiao, L. (2025). Stochastic Approximation with Block Coordinate Optimal Stepsizes. arXiv preprint arXiv:2507.08963.
We derive adaptive block-coordinate stepsizes from online second-moment estimates, matching Adam's empirical performance with less memory and fewer hyperparameters.

II. AI for Science and Science for AI

Lotfi, S., Jiang, T., Saxena, E., Raugel, J., Cascardi, E., Gromov, A., King, J.-R., Williams, A., & Lauter, K. (2026). Brain Data as a Block-Importance Metric for Depth Pruning. (Working paper)
We explore whether correlations between brain data and AI models can indicate block importance and design a depth-pruning algorithm based on brain-model alignment scores.

III. Optimization Algorithms and Theory

Jiang, T., Moursi, W., & Vavasis, S. (2023). Range of the displacement operator of PDHG with applications to quadratic and conic programming. arXiv:2309.15009.
We characterize the range of the PDHG displacement operator using monotone operator theory. The result shows that PDHG can diagnose infeasible or unbounded quadratic programs and ellipsoid-separation problems.

Davis, D., & Jiang, T. (2022). A linearly convergent Gauss-Newton subgradient method for ill-conditioned problems. arXiv:2212.13278.
We prove linear convergence of a Gauss-Newton preconditioned subgradient method for composite objectives under constant-rank, semismoothness and sharpness assumptions. Its oracle complexity is invariant under smooth reparameterizations.

Jiang, T., Wang, S., Zhang, R., Qin, L., Wu, J., Wang, D., & Ahipasaoglu, S. D. (2019). An inexact ℓ_2-norm penalty method for cardinality constrained portfolio optimization. *The Engineering Economist*, 64(3), 289297. doi: 10.1080/0013791x.2019.1636169
We study portfolio optimization with nonconvex constraints on sector weights and the number of holdings. A reformulation and inexact ℓ_2-norm penalty method simplify the resulting computation.

IV. Machine learning: mathematics for clustering

Jiang, T., Tan, S., & Vavasis, S. (2023). Re-embedding data to strengthen recovery guarantees of clustering. arXiv preprint arXiv:2301.10901.
We propose leapfrog distances to re-embed data before applying sum-of-norms clustering. We prove that this pipeline strengthens guarantees for recovering the underlying cluster structure.

Jiang, T., Vavasis, S., & Zhai, C. W. (2020). Recovery of a Mixture of Gaussians by Sum-of-norms Clustering. Journal of Machine Learning Research, 21(225), 1-16.
We remove the sample-size upper bound from recovery guarantees for sum-of-norms clustering. The method can recover Gaussian mixtures even as the number of samples tends to infinity.

Jiang, T., & Vavasis, S. (2020). Certifying clusters from sum-of-norms clustering. arXiv preprint arXiv:2006.11355.
We propose a test that identifies and certifies exact cluster assignments from approximate primal-dual solutions of sum-of-norms clustering. A primal-dual path-following algorithm is guaranteed to produce a certifiable solution after sufficiently many iterations.

SERVICES

Reviewer for ICML, SIAM Journal on Optimization, SIAM Journal on Mathematics of Data Science, JMLR, Machine Learning
Co-chair of the minisymposium *Mathematics for sum-of-norms clustering* at the 2022 SIAM Conference on Mathematics of Data Science

HONORS AND AWARDS

Cornell: Eleanor and Howard Morgan PhD 68 Graduate Fellowship
UW: William Tutte Postgraduate Scholarship, Sinclair Graduate Scholarship, MATH Senate Graduate Scholarship, UW Graduate Scholarship, TA and RA fellowships
Ministry of Education, Singapore: SM2 scholarship
SUTD: SRX Operations Research Award, top student in Operations Research, SUTD Honors List for all academic terms

TEACHING EXPERIENCES

Graduate Teaching Assistant, Cornell University, USA

ORIE 5380 Optimization Methods	*2022 Fall*
ORIE 4740 Statistical Data Mining	*2022 Spring*
ORIE 4741 Big Messy Data	*2021 Fall*
ORIE 3310 Optimization II	*2021 Spring*